Integration: Substitutions Mix

Questions 1–8

Question 1. For each of the following, suggest a substitution for x that reduces the integral to $\int C\,d\theta$, where C is a constant.

a) $\displaystyle\int \frac{1}{\sqrt{a^2-x^2}}\,dx$, for constant a

b) $\displaystyle\int \frac{1}{\sqrt{a^2+x^2}}\,dx$, for constant a

c) $\displaystyle\int \frac{1}{a^2-x^2}\,dx$, for constant a

d) $\displaystyle\int \frac{1}{\sqrt{x^2-a^2}}\,dx$, for constant a

e) $\displaystyle\int \frac{1}{a^2+x^2}\,dx$, for constant a

Question 2. For each of the following integrals, suggest a suitable substitution and then integrate.

a) $\displaystyle\int \frac{1}{\sqrt{9x^2+400}}\,dx$

b) $\displaystyle\int \frac{1}{\sqrt{9-16x^2}}\,dx$

c) $\displaystyle\int \frac{1}{36-x^2}\,dx$

d) $\displaystyle\int \frac{1}{\sqrt{625x^2+1}}\,dx$

e) $\displaystyle\int \frac{1}{\sqrt{121x^2-16}}\,dx$

f) $\displaystyle\int \frac{1}{1+25x^2}\,dx$

g) $\displaystyle\int \frac{1}{\sqrt{100x^2+9}}\,dx$

h) $\displaystyle\int \frac{1}{\sqrt{4-121x^2}}\,dx$

i) $\displaystyle\int \frac{16}{\sqrt{x^2-4}}\,dx$

j) $\displaystyle\int \frac{7}{144+49x^2}\,dx$

k) $\displaystyle\int \frac{24}{25-9x^2}\,dx$

l) $\displaystyle\int \frac{32}{\sqrt{81-2x^2}}\,dx$

m) $\displaystyle\int \frac{90}{\sqrt{9x^2-10}}\,dx$

n) $\displaystyle\int \frac{55}{50-121x^2}\,dx$

o) $\displaystyle\int \frac{36}{5+108x^2}\,dx$

Question 3. For each of the following integrals, suggest a suitable substitution and then integrate.

a) $\displaystyle\int \frac{3}{\sqrt{9x^2+4}}\,dx$

b) $\displaystyle\int \frac{14}{4+49x^2}\,dx$

c) $\displaystyle\int \frac{144}{9-16x^2}\,dx$

d) $\displaystyle\int \frac{8}{\sqrt{16x^2-9}}\,dx$

e) $\displaystyle\int \sqrt{25-4x^2}\,dx$

f) $\displaystyle\int \frac{24576}{(16-9x^2)^3}\,dx$

g) $\displaystyle\int (4-49x^2)^{3/2}\,dx$

h) $\displaystyle\int \sqrt{36x^2+25}\,dx$

i) $\displaystyle\int \frac{900000}{(100+9x^2)^3}\,dx$

j) $\displaystyle\int \frac{280}{(25-49x^2)^2}\,dx$

k) $\displaystyle\int \frac{250}{(25+2x^2)^2}\,dx$

l) $\displaystyle\int (3x^2-4)^{3/2}\,dx$

Question 4. For each of the following integrals, suggest a suitable substitution and then integrate.

a) $\displaystyle\int \frac{1}{\sqrt{1-(3x-1)^2}}\,dx$

b) $\displaystyle\int \frac{1}{\sqrt{(x-1)^2+9}}\,dx$

c) $\displaystyle\int \frac{1}{\sqrt{3-(2x+5)^2}}\,dx$

d) $\displaystyle\int \frac{1}{\sqrt{(3x-2)^2+8}}\,dx$

e) $\displaystyle\int \frac{1}{8+(4x-1)^2}\,dx$

f) $\displaystyle\int \frac{4}{11-(5x+3)^2}\,dx$

Question 5. For each of the following integrals, suggest a suitable substitution and then integrate.

a) $\displaystyle\int \frac{1}{4x^2+12x+10}\,dx$

b) $\displaystyle\int \frac{3}{\sqrt{3-4x-4x^2}}\,dx$

c) $\displaystyle\int \frac{2}{x^2-6x+14}\,dx$

d) $\displaystyle\int \frac{1}{\sqrt{7+4x-x^2}}\,dx$

e) $\displaystyle\int \frac{4}{3x^2+8x+6}\,dx$

f) $\displaystyle\int \frac{7}{\sqrt{5x^2+2x+12}}\,dx$

Question 6. By first suggesting a suitable substitution, show that:

a) $\displaystyle\int_0^1 \frac{2}{\sqrt{5-(2x-1)^2}}\,dx = 2\arcsin\left(\frac{\sqrt{5}}{5}\right)$

b) $\displaystyle\int_0^2 \frac{8}{\sqrt{4+(x-1)^2}}\,dx = 8\ln\left(\frac{3+\sqrt{5}}{2}\right)$

c) $\displaystyle\int_{-3}^{-3/2} \frac{3}{45+24x+4x^2}\,dx = \frac{\pi}{8}$

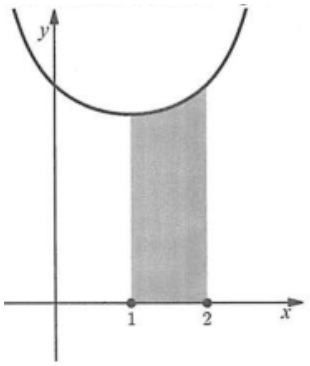
d) $\displaystyle\int_0^{3/2} \frac{2}{\sqrt{4x^2-12x+15}}\,dx = \ln\left(\frac{\sqrt{2}}{\sqrt{5}-\sqrt{3}}\right)$

e) $\displaystyle\int_{-1}^{4} \sqrt{x^2+2x+10}\,dx = \frac{135}{32}+\frac{9}{2}\ln 2$

Question 7. The curve shown in the diagram below has equation

$$y = \frac{5}{\sqrt{3 + 2x - x^2}}.$$

Find the exact area of the shaded region.



Question 8. The curve shown in the diagram below has equation

$$y = x + \frac{4}{6 + 4x + x^2}.$$

Find the exact area of the shaded region.

